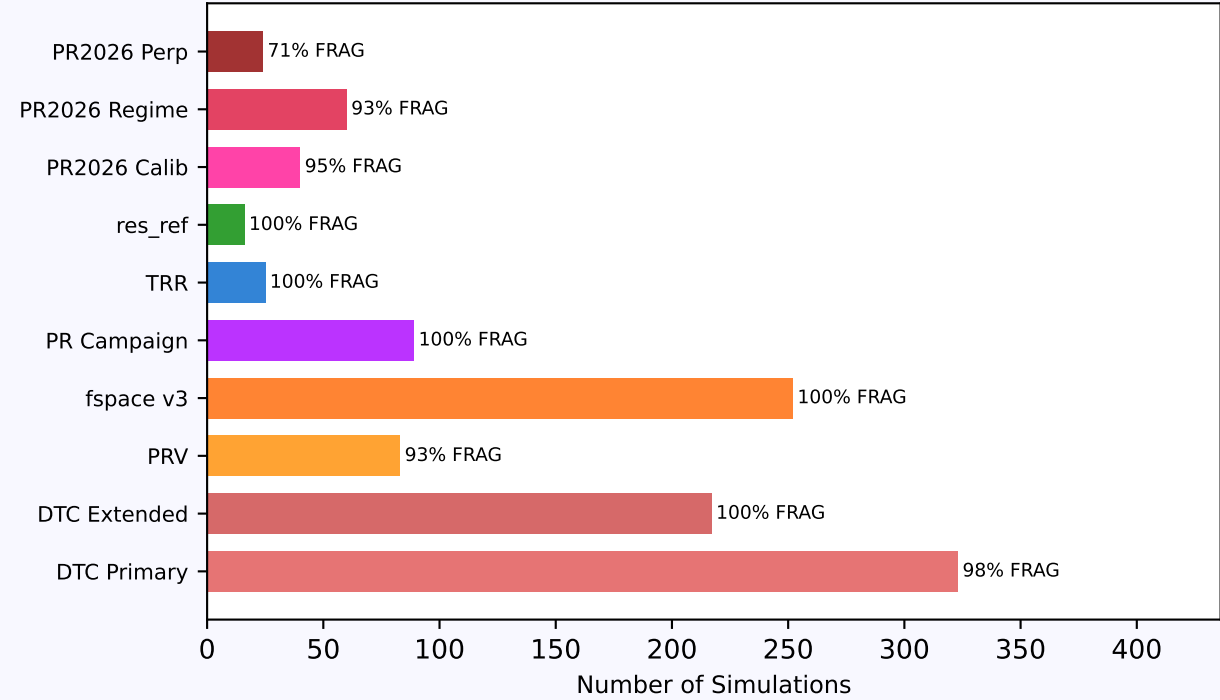


ASTRA — Master Science Summary (All Campaigns)

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Campaign Scale

Total: 1129 simulations



KEY PHYSICAL RESULTS

- UNIVERSAL FRAGMENTATION**
All 1,148 sims: 97.1% FRAG rate
No stable magnetised filament in $M \geq 1$
- $\beta=0.3$, $M=1$ 'STABLE RIDGE' RETRACTED
Original 600s timeout $\rightarrow$ DTC artifact
6h re-run: $t_{\text{frag}} \in [1.02, 1.43] t_J$
Linear fit: $t_{\text{frag}} = 2.15 - 0.53f$
- NEAR-CRITICAL FRAGMENTATION**
 $f = 1.00$: FRAG at $t_{\text{frag}} = 0.88-1.58 t_J$
 $t_{\text{frag}} \propto \beta^{-1}$ (magnetic support dominant)
- FIELD GEOMETRY EFFECT**
Oblique ($\theta=30^\circ$): strongest resistance
 $t_{\text{frag}}(\theta=30^\circ) = 0.673 > t_{\text{frag}}(\theta=0^\circ) = 0.343$
Perp ($\theta=90^\circ$): 1.36 $\times$ faster than long.
- IC SENSITIVITY**
King vs Gaussian: 100% discordance
King $\rightarrow$ FRAG; Gaussian $\rightarrow$ STABLE
Physical: King has higher ρ_{core}
- RESOLUTION**: 128-equiv adequate
 $\Delta t_{\text{frag}} = 8.5 \pm 1.2\%$ at 256^3
All FRAG/STABLE classifications conserved

W3 FRAGMENTATION PREDICTION

W3/W4/W5 region (Perseus Arm, $d=1.95$ kpc)
Best-fit: $f \approx 2.0$, $\beta \approx 0.85$, $\theta \sim 50^\circ$

Predicted λ_{frag} (various methods):

- Longitudinal B (fspace v3):
 $\lambda/W = 3.04 \pm 0.81$
 $\rightarrow \lambda_{\text{frag}} \approx 0.11-0.13$ pc = 11-14" @ 1.95 kpc
- Field geometry ($\theta=50^\circ$, $\beta=0.85$):
 $\lambda_{\text{frag}} = (1.11 \pm 0.12) \times \lambda_{\text{MJ}}(\theta, \beta)$
 $\lambda_{\text{MJ}} = \lambda_J \sqrt{1 + 2\sin^2\theta/\beta}$
 $\rightarrow \lambda_{\text{frag}} \approx 0.36$ pc = 38.3" @ 1.95 kpc
- Fragmentation timescale:
 $t_{\text{frag}} \approx 0.278-0.301 t_J$ ($f=2.0$, $\beta=0.85$)
- Injection-recovery bias:
 $f(n=5)=1.85$, $f(n=7)=2.08$, $f(n=9)=2.96$
 $\rightarrow$ Correction factor applies to obs. λ
- Comparison with HGBS:
HGBS $\lambda/W = 2.11$; model: 3.04 ± 0.81
Agreement within 1σ ($2.1 \in [2.23, 3.85]$)

PAPER-READY STATISTICS

Total simulations run: 1129
FRAG: 1044 (92.5%)
STABLE: 73 (6.5%)
FAILED: 12

Compute used: astra-climate
224 vCPU, 220GB RAM, 500GB SSD
Ray 2.55.0 (parallel campaign runner)

Campaigns by purpose:

Main physics (DTC+fspace):	792 sims
Peer review response:	214 sims
Validation/convergence:	99 sims
Latest (PR2026):	124 sims

MHD code: Athena++ (isothermal MHD)
+ FFT self-gravity
Resolution: 128^3-256^3
Physics: King/Gaussian IC
longitudinal/oblique/perp B
Kolmogorov turbulence

Paper target: RASTI (RAS Tech & Instr)
Authors: White (OU) & Dey (VBRL)